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Speech Recognition Lab

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Abstract

Automatic speech recognition (ASR) in the military domain can enable voice user interfaces of command & control information systems (C2IS) and, thus, improve the usability of C2IS. It can also serve intelligence (COMINT) by supporting the monitoring of radio communication. However, military use cases are challenging as ASR has to cope with adverse conditions due to speech variations, noise and other channel disturbances, as well as strong requirements regarding reliability and robustness. Generic ASR solutions do not meet these requirements sufficiently. Therefore, tailor-fit ASR solutions have to be provided. In order to enable the quick development and testing of such solutions for specific use cases, we designed a Speech Recognition Lab (SR Lab). A demonstrator of the SR Lab and its components has been implemented and will be described in this paper.

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1. Introduction

Command & control information systems (C2IS) are used at various command levels, from the strategic down to the tactical, where they are applied, e.g., in vehicles such as battle tanks. There, challenges arise in terms of operability. C2IS must be operable under adverse conditions when soldiers are exposed to stress and occupied with higher-priority tasks, so that they cannot focus on interacting with the system. Compensation must also be made for noise, poor visibility, or other disturbances. To ensure operability, the application of automatic speech recognition (ASR) is an obvious solution. However, ASR must function reliably under the adverse conditions mentioned [17]. Other military applications for ASR, e.g., in the field of radio reconnaissance and surveillance, are also characterized by adverse conditions: different microphone qualities, distorted signals, e.g., due to transmission over an RF-channel, changing speech or speaker behavior, as well as high demands on robustness. ASR systems must be adapted to these conditions.

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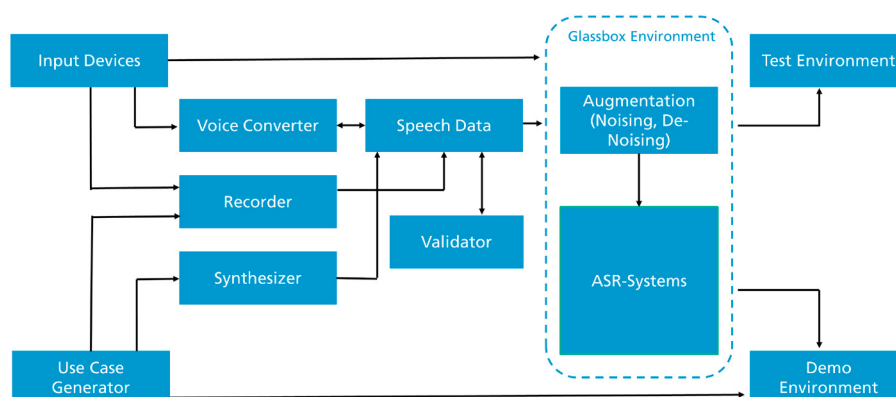


Fig. 1: SR Lab Architecture.

We distinguish two kinds of ASR systems, namely large vocabulary continuous speech recognition (LVCSR) systems, which fully transcribe arbitrary speech input, and small footprint keyword spotting (KWS) systems, which recognize only selected keywords and phrases. If a language fragment is, by domain knowledge, known to belong to a restricted set of possible utterances, it is in several cases possible to cast the ASR into a keyword-spotting task. Commercial ASR systems that we encounter in everyday life are usually LVCSR systems that have been trained with very large amounts of data (several thousands of hours of transcribed audio material). However, they have not been optimised for the adverse conditions of military use cases. Data volumes such as those needed for training commercial ASR systems are usually not available for training systems in the military sector.

Military use cases are thus characterised, firstly, by the fact that ASR must function robustly under adverse conditions. Secondly, transcribed training data for ASR systems are only available to a limited extent, and the generation of additional training data is very costly. Thirdly, however, for military use cases, it is not necessary to capture natural language, like German or English, completely, but only a subset depending on the particular application.

The aim is to tailor and optimise ASR systems for defined military use cases, their specific contexts of use and individual users. It should also be possible to quickly train and deploy ASR systems for new use cases, i.e., new language fragments. The task we face is therefore not to construct a generic ASR system, but to provide an environment with which specific speech recognisers can be constructed as needed. We call such an environment a “Speech Recognition Lab” (SR Lab). Within this paper, we will describe the architecture and a demonstrator of an SR Lab and its components.

This is the outline of the paper: in Section 2, we will explain the overall architecture of the SR Lab and its components. In Section 3, we will sketch two different military use cases, namely speech interaction with an integrated C2IS and monitoring of radio communication for transcribing transmissions of coordinates. Then, we will describe two ASR systems that are part of the SR Lab. Those two are a KWS system (Section 4) and an LVCSR system (Section 5). In the final Section 6, we will summarize our work and give an outlook on future work.

2. Speech Recognition Lab

The SR Lab is intended to serve several purposes: firstly, it serves as an environment for conducting ASR experiments. Among other things, it should enable competing solutions to be evaluated and compared with each other. In this way, it is to be determined which requirements are already fulfilled by existing solutions and for which use cases new systems or further improvements of existing systems are needed. In addition, the SR Lab is intended as a demonstration environment to support the explanation of different processes and their potentials. The aim is to make transparent and give operators an understanding of different approaches’ strengths and limitations. Finally, as a productive environment, the SR Lab should enable the generation of new ASR solutions on demand.

Figure 1 depicts the general architecture of the SR Lab. Most of its components have already been implemented in our demonstrator. However, some are still to be realised and others have to be further developed.

Speech data, recorder and validator: speech data is required for training and testing ASR systems. For this purpose, freely available or commercial datasets can be accessed. However, such datasets do not cover the specific requirements of military use cases, which is why we made recordings both in simulated conditions and under real conditions in battle tanks as well as in infantry fighting vehicles. In addition, we implemented a recorder with a web interface. It allows users to record their own data so that personalised models can be trained. To do this, the user selects a dataset and successively records items until all the required data is available. The user can interrupt recording at any time and continue later. Two options are available for quality assurance: firstly, each user can listen to her recording again and correct it if necessary. Secondly, all recordings of all users can be listened to and evaluated one after the other via a validator interface.

Input devices: we can record data with different microphones, namely a conventional pressure microphone in the form of a headset, a throat microphone and a skullcap microphone. The recordings can be fed directly into a speech recognition system. When using the recorder, it is possible to select which microphone is used and thus make microphone-specific recordings. For recordings under real conditions in the tank, we have connected a recording device to the on-board communication system.

Use case generator: so far, we have specified our use cases manually by defining a language fragment in the form of a vocabulary and a grammar (cf. Section 3). Based on vocabulary and grammar, we have generated 1-, 2-, 3-, 4- and 5-grams for training a language model. We have also generated exemplary training and test sentences. To enable users to easily generate new use cases, a use case generator is to be created. It will have an intuitive interface that guides the user through the process of use case definition, and also enables the automatic generation of N-grams and sentences. N-grams are needed to train a language model. Sentences are needed, e.g., to make scripted recordings for training an audio classifier.

Synthesizer and voice converter: training and test data do not necessarily have to be recorded by a human speaker, a process requiring a lot of effort and making it difficult to adapt quickly to new use cases and speakers. Rather, such data can also be generated automatically. For this purpose, text-to-speech (TTS) synthesisers are available, which can be used to acoustically realise new texts for given speakers. The problem here, however, is the naturalness of the synthesised speech and thus its suitability for training. Secondly, voice conversion (VC) can be used to transfer given speech data into the voices of other speakers. Recordings of one speaker are manipulated in such a way that they sound like recordings of another speaker. We have conducted experiments on voice conversion to see how well it can be used to personalise an ASR system. A voice converter includes a synthesiser. One of the future steps will consist in also adding a general TTS component into the SR lab in order to complement the VC capabilities.

Automatic speech recognition systems: we have implemented two different ASR systems, namely a KWS and an LVCSR system. Both are described in Sections 4 and 5, respectively. To have benchmarks for comparisons, we have also conducted experiments with commercial systems.

Audio manipulation: before a signal is passed to a speech recogniser, it can be manipulated either to improve recognition – e.g. by filtering/reducing noise – or to simulate special conditions of use. For the latter, specific noise can be added or the bandwidth can be reduced to simulate radio conditions or similar. Audio manipulation/augmentation, thus, also serves the creation of training data. (Cf. Section 4.)

Testing environment: evaluation measures have to be defined in order to compare different techniques and models. These measures strongly depend on the respective application and the user requirements. So far, the testing environment is merely a set of evaluation measures. We are planning to automate the evaluation process and provide the SR Lab user with an interface that guides her through the evaluation process.

Demonstration environment: the experimental C2IS demonstrator “Intelligent Situational Awareness” (InSAne, [6]), developed at Fraunhofer FKIE, serves as a demonstration environment. InSAne enables the rapid development and integration of micro services. This allows, e.g., to prove the value of an ASR-driven voice user interface (VUI) for the definition and update of battle space objects (BSO). To this end, a dialogue manager has been implemented that updates a situational picture according to an operator’s speech input. (Section 3.1.)

Glassbox environment: the aim of the glassbox environment is to serve as a tool for the operator to assess the capabilities of the various ASR related modules used in the SR lab environment. In particular, analysis steps, generated hypotheses and the selection of a hypothesis are to be presented and made comprehensible. The purpose is to promote understanding of the various techniques and thereby guide the selection of a system or its targeted modification. Examples (screenshots) of respective visualisations will be given in Sections 4 and 5.

3. Military Use Cases

We have investigated two different use cases for the application of ASR in the military sector, namely a VUI to an integrated C2IS and the automatic recognition of coordinates that are transmitted via radio. Both use cases have been defined for the German language. The SR Lab has been applied for respective data collection as well as the development and evaluation of ASR solutions.

3.1. Voice User Interfaces of Integrated C2IS

Modern tanks and other military vehicles are equipped with integrated C2IS. The aim is to provide the crew with a common map-based situational picture. The crew itself contributes to the situational picture with information on newly detected BSOs or updates on BSOs that have been seen already. It has been shown that the operation of an integrated C2IS via a touchscreen and a keyboard is practically impossible in situations of high intensity [17]. In such situations, an operator cannot focus her attention on the system. It is precisely then, however, that updating and supplementing the existing information would be of particular importance. If this is not done, the situational picture loses its topicality and completeness.

A promising solution is to complement the given graphical user interface (GUI) with a VUI by which the operator can create new BSOs in the system or update existing BSOs. The development of a VUI for use in tanks and other vehicles is on the one hand a challenging and on the other hand a rewarding task. It is challenging because, firstly, there are varying noise levels that interfere with the speech signal. Secondly, tanks are subject to constant vibrations, which cause the recording quality to change continuously. Thirdly, the crew may be exposed to stress, which affects speech and interaction behaviour. Under stress, the fundamental frequency, volume and speaking rate typically increase, while articulation accuracy may decrease. This presents a challenge for ASR. Finally, both the operator's ability and willingness to adapt to a complex system and to compensate or correct for errors decrease. A VUI must therefore be particularly easy to use and also function robustly under the expected adverse conditions.

We have defined a dialogue model for the interaction with an integrated C2IS. It enables the following speech-based actions:

1. Creation of a new BSO: for creating a new BSO, at least its type has to be named. Possible types are, among others, "Kampfpanzer" (battle tank), "Schützenpanzer" (infantry fighting vehicle), "Drohne" (drone, UAV) etc. The location of the BSO can be automatically specified by targeting the respective object using a laser. Further attributes can either be named before the BSO type – "feindlicher Kampfpanzer" (hostile battle tank) – or after it – "Kampfpanzer, feindlich" (battle tank, hostile). Contrary to English, German requires morphological marking in order to assure syntactic agreement. Therefore, different morphological forms of the same word ("feindlich", "feindlicher", "feindliche", "feindliches") have to be included. Attributes can be simple words or complex phrases consisting of several words: "Vermutlich neutraler LKW, in Bewegung nach Süden." (Probably neutral truck, moving south.) Finally, new BSOs can be named so that an operator can repeatedly refer to the same BSO by using its name. To this end, we allow phrases like "Name: Brady 1". By standard, we define names as consisting of the words "Brady" or "Generic" for hostile or friendly BSOs, respectively, and a number. A name is only valid for the operator who assigned the name to the object, so that there will not be conflicts arising from different operators assigning the same name to different objects.
A complex sentence for creating a new BSO can thus be: "Feindlicher Kampfpanzer, in Stellung, Ausrichtung nach Nordwesten, Name: Brady 1." (Hostile battle tank, in position, orientation northwest, name: Brady 1.)
2. Selecting an existing BSO: before a BSO can be updated, it has to be selected, either by selecting it on the map or by saying its name, e.g., "Brady 1".
3. Updating an existing BSO: every sentence that can be used for creating a new BSO can also be used for updating an existing BSO. However, updates do not require the naming of a BSO type, which is mandatory in the creation of a new BSO. A very short update would be "zerstört" (destroyed).
4. Transmission: newly created or updated BSOs can be transmitted to other C2IS: "Versende, Sendekreis Eins" (Transmit to radio circuit 1.)

5. Dialogue control: it must be possible to correct an input, and abort or successfully finish a dialogue. Respective keywords for dialogue control are “Korrigiere” (correct), “Abbruch” (abort) und “Eingabeende” (end of input), among others.
6. Alarms and Warnings: also standardized alarms and warnings, e.g. mine warnings, are part of the language fragment.
7. GUI control: finally, interaction with the map view can be simplified by including commands like “zoom in”, “refresh”, “renew”, etc. (in English and German).

Except for creating and updating BSOs, all speech acts are of very low complexity. They can be defined by merely enumerating the possible sentences. Defining BSO inputs, however, requires a vocabulary and a grammar.

In total, the vocabulary for all speech acts includes 150 base forms (lemmas) and 195 word forms, including morphological variants. This is a large vocabulary for a KWS system. It rather favours the application of an LVCSR system.

Among the words of the vocabulary are 58 expressions that denote BSO types. Further words serve the generation of phrases that refer to attributes. In principle, it is possible to assign arbitrarily many attributes to a BSO. However, to exclude meaningless or contradictory sentences like “Tank moving south, moving west”, attributes have been assigned to different categories. Sentences for creating or updating BSOs may contain only one attribute from each category. The grammar was thus defined in such a way that it excludes semantically meaningless sentences as far as possible. The defined language is nevertheless large. For training a language model, 195 unigrams, 4,230 bigrams, 86,499 trigrams, 1,307,243 4-grams and 16,901,930 5-grams can be formed. The number of possible sentences is many times higher. The vocabulary and grammar can be easily modified so that the model can be adapted to new application scenarios where other types of BSOs are expected and likely to be entered into the C2IS.

For training ASR systems, we made use of the recorder and the validator to gather audio data on the base vocabulary, the base phrases denoting individual attributes and entire sentences of the language fragment. In total, we recorded and validated more than 19 hours of audio material. Furthermore, we made scripted recordings in a tank simulator and in tanks during exercises. That way, we were able to generate approx. one hour of additional material (including material for the use cases described in the next subsection). Obviously, these are very few data samples compared with the thousands of hours of audio material available for generic uses cases in the non-military sector. The challenge is to solve the apparent scarce data problem.

3.2. Monitoring of Radio Communication

The second use case is the detection and recognition of coordinates transmitted by radio to a command post. In command posts, coordinates are not always understood and it is often necessary to ask for repetition and reconfirmation. Comprehension problems can be caused acoustically (for example, by noise in the vicinity of the person reporting), but can also be due to a drop in concentration or high speaking rates. Even requests for repetition cannot ensure correct transmission. In order to facilitate the work in the command post and to help avoiding mistakes, we propose the use of a KWS system for the automated recognition of coordinates mentioned over the radio. The system has the task of monitoring the radio and displaying named coordinates in written form. Detected coordinates can also facilitate the control of a C2IS, e.g., by automatically adjusting the map section displayed and marking the coordinate in it.

The problem of recognising coordinates in radio messages only exists as long as coordinates are reported by radio at all. The problem does not arise if BSOs including coordinates are sent as data packets. However, it cannot be assumed that radio messages with coordinates will be completely dispensed in the foreseeable future. In this respect, coordinate detection is a practical use case for ASR.

Information transmitted by radio to a command post are in a standardized format. In German, coordinates are composed of numbers, the NATO alphabet (for reference points), “plus” and “minus” and markers for correction like “(Ich) korrigiere” (I correct): “Bravo Minus Sieben Sechs Plus Zwo Vier Korrektur Minus Null Sechs.” (Bravo minus seven six plus two four correction minus zero six.) Due to the small vocabulary, it seems feasible that a robust system can be trained and deployed despite degraded audio quality due to transmission over radio.

We recorded and validated approx. seven hours of respective audio material with the recorder and the validator of the SR Lab. In addition, few data samples have been recorded during exercises in tank simulators and tanks (cf. the preceding subsection).

Coordinate detection can be seen as a special and manageable case of radio monitoring. It can be modified in various ways. There are a variety of operational scenarios in the field of communications intelligence (COMINT). They require that words in different languages can be recognised. Coordinate recognition can also be extended to recognise entire radio messages, which then can lead to the automatic updating of the situational picture. The information units for such a use case are to a large extent already included in the VUI and the coordinate recognition use cases. For the recognition of radio messages, the two use cases would have to be merged. However, this would make the vocabulary quite large for a KWS system.

4. Keyword Spotting

Now, we will describe the development and evaluation of a keyword spotting (KWS) system for coordinates recognition, using the SR Lab environment. A description of the system including all technical details can be found in [20].

4.1. Audio manipulation

The creation of the dataset of the use case has already been described in Section 3. However, for the development of a KWS system the audio files contained in the dataset need to be pre-processed appropriately. This is done with the audio manipulation component of the SR Lab.

The first stage of audio manipulation is the extraction of suitable speech features to reduce the dimension of the data. This is important because raw audio waveforms are very high-dimensional containing much redundant information. To train an ASR system with waveforms, a very large amount of data is needed that in many cases, especially in the military domain, is not available. When reducing the dimension of the data first while preserving all relevant information, less data is needed. For this purpose, log-Mel spectrograms, which are time-frequency representations computed by applying a logarithmically scaled triangular filterbank (Mel filterbank) to the short-time Fourier transformation of the waveform, commonly used in many audio classification tasks including speech recognition, are chosen.

Before trying to recognize speech, it is helpful to first detect time frames that contain any speech in order not to recognize spoken words where only noise is present. This step is called voice activity detection (VAD). For this purpose, the spectral energy in two pre-defined frequency subbands as well as spectral flux and spectral flatness are monitored over time [13, 18, 12]. By comparing these features to a threshold that is empirically tuned, regions of actual speech are detected.

Since the dataset containing the individual keywords available in our use case is very small, data augmentation is applied while training in order to generate additional data and avoid overfitting of the system. Without this step, the system would only remember the training samples without learning anything about the underlying structure leading to bad generalization capabilities when being used on previously unseen data samples. More concretely, a version of manifold mixup [4] and SpecAugment [2] are used. For manifold mixup, regular mixup [22], which generates random linear interpolations between training samples for training the model, is applied to every intermediate representation inside a neural network (including the input). SpecAugment itself consists of three different data augmentation techniques, namely time warping, time masking and frequency masking, and is only applied to the input. For time warping, one part of the spectrogram of random size is stretched whereas the remaining part is shrunk appropriately. For time masking and frequency masking, random consecutive time steps or frequency bins are replaced with zeros.

In order to train a KWS system, one needs strongly labeled data, i.e. isolated recordings of keywords. Obtaining such data is very costly and requires experts to manually look for these keywords and to listen multiple times until their exact start- and endpoints are determined. It is much less costly to only annotate which words occur without the need to determine their precise location in time because in this scenario experts usually only need to listen once to do this. A dataset labeled this way is called weakly labeled. To save human resources, strong labels for a dataset consisting of weakly labeled sentences are generated by using dynamic time warping (DTW) with so-called HFCC-ENS features [19].

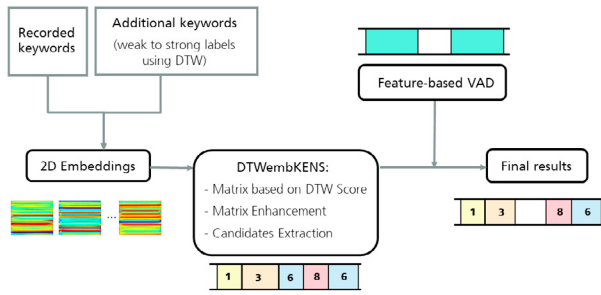


Fig. 2: Overview of the KWS system.

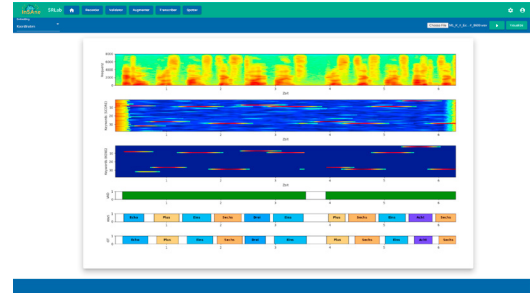


Fig. 3: Exemplary illustration for a recorded sentence.

4.2. ASR module

The KWS system itself consists of three major stages. First, embeddings are extracted by training a neural network to discriminate among the keywords. Second, dynamic time-warping (DTW) is applied. Third, a post-processing based on certain *similarity matrices* computed during the DTW step is applied to obtain start- and endpoints of keyword detections. An overview is given in Figure 2.

To obtain representations of the time-frequency features that are specifically trained to highlight information for discriminating among the keywords, a convolutional neural network in combination with a two-dimensional version of the adaptive angular margin loss AdaCos [23] has been used resulting in new 2D feature embeddings. Unlike their one-dimensional counterpart, these embeddings have the advantage that one of their axes can be interpreted as time. Therefore, they can represent speech signals of varying lengths and thus be used to search for keywords with DTW. All training samples of the same keyword are combined into a single pattern that can be searched later by using the DTW barycenter averaging algorithm [16] as implemented in [3]. The similarity of each of these patterns to segments of a given audio file is calculated by comparing each feature vector, interpreting each pattern as a sequence of feature vectors. Based on these similarities, accumulated cost matrices are calculated using DTW. However, instead of extracting warping paths, only the last row of each accumulated cost matrix is considered and concatenated in a so called score matrix, where high values indicate high similarity. The score matrix is then enhanced using normalization, quantization, smoothing, and downsampling – steps based on a technique from music processing to calculate chroma energy normalized statistics (CENS) features [15, 14]. The last step is to extract the final candidates by picking the maxima of the enhanced matrix (keyword energy normalized statistics (KENS) matrix). Domain specific knowledge and the results of the VAD algorithm are finally included to discard false positive detections. For more details, the reader is referred to [20].

4.3. Glassbox environment

The glassbox environment is helpful to understand the different steps included in the speech recognition process of the KWS system. An example is shown in Figure 3. The most important steps of the system are shown as extra plots. The first plot is the spectrogram of the selected audio file, the second and third plots illustrate the score matrix and its enhanced form, the KENS matrix, respectively. The VAD results can be seen in the plot below as green areas. The rectangles in the second to last row are the recognized keywords of the KWS system. Each keyword has a different color, in order to facilitate the comparison with a (possibly) available ground truth, given in the last plot.

4.4. Evaluation

For this particular use-case, the word-error-rate (WER) was chosen as a metric to measure the performance of the KWS system. To calculate the WER, an automatically predicted sequence of words is compared to the correct

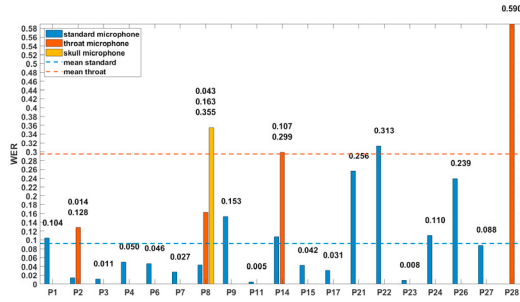


Fig. 4: Exemplary evaluation of the KWS system.

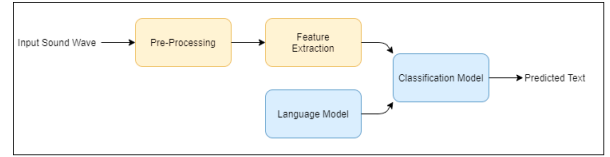


Fig. 5: Overview of the LVCSR system.

sequence of words that had to be predicted by the ASR system. The WER is defined as

$$\text{WER} := \frac{S + D + I}{S + D + C} \quad (1)$$

where S denotes the number of substituted words, D denotes the number of deleted words, I denotes the number of wrongly inserted words and C denotes the number of correctly recognized words.

The evaluation of the KWS system is depicted in Figure 4. The x-axis denotes different speakers and the y-axis the WERs resulting from all test recordings of these speakers. Depending on the type of microphone, differently colored bars are used to visualize the individual WERs obtained with all data belonging to these microphones. Additionally, the means of the WERs from all speakers obtained with all but the skull cap microphone types are depicted with dashed horizontal lines. (As only one speaker made recordings with the skull cap microphone, the computation of a mean value seems inappropriate.) Using this graphical evaluation, one can quickly identify speakers that are recognized well and speakers that are not and compare the average performances of different microphone types.

5. Large Vocabulary Continuous Speech Recognition

This section describes the large vocabulary continuous speech recognition (LVCSR) module for the SR Lab environment, as shown in Figure 5. The LVCSR module defines the recognition task as a sequence to sequence problem. A continuous stream of audio input is received, then the corresponding characters are produced as the output. In the cases of “out of vocabulary” recognition tasks, a predicted sequence of characters would yield an expected text. Yet, the module also utilizes a language model component to optimize the recognition results for the defined use cases. The following subsections explain the relevant details of the LVCSR module.

5.1. Audio manipulation

This section focuses on the pre-processing and feature extraction blocks from Figure 5. The dataset collection is an essential part of a data-driven system. Yet the preparation of the collected data to obtain a high-quality dataset for the training step is also crucial. Therefore, the LVCSR module utilizes the audio manipulation module, as it has been described in Section 4.1, except for the strong label generation case. Since the expected data volume is larger for LVCSR training, the CTC-loss function [9] is selected accordingly to cope with only weak labels, aiming to minimize the manual validation effort.

5.2. Language model generation

The task of a language model in a recognition pipeline is to reduce the error on the noisy and ambiguous inputs. Predicting the correct word in a given sequence supports solving this problem. This probability can be estimated

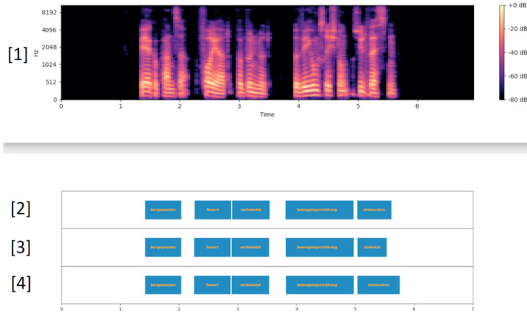


Fig. 6: Exemplary illustration for a recorded sentence.

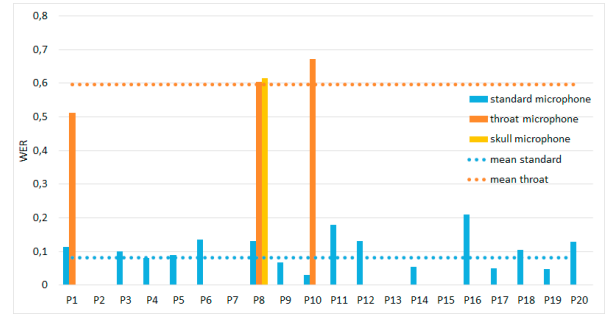


Fig. 7: Exemplary evaluation of the LVCSR system.

using maximum likelihood estimation. The N-gram probability is estimated by dividing the considered sequence's frequency by the currently observed prefix's frequency. This naive intuition has a flaw in case of unknown sequences in the training corpus. In the designed language model, this issue is solved by introducing a back-off mechanism as described in [11] and using an optimized implementation as described in [10]

5.3. ASR module

In the case of the ASR, an acoustic model is trained in combination with the described audio manipulation tools and the language model. The acoustic model, inspired by [5] and further investigated in [7], is a combination of a convolutional and a recurrent neural network. Each data sample from the dataset contains an utterance x and the corresponding label y . Every single utterance is a time-series where each slice is a vector of audio features as described in the audio manipulation section (4.1). The task of the network is to convert the input sequence to the expected transcript. Generated predictions of the network are the graphemes of the target language. In each step, the network produces a prediction over those characters and is trained discriminatively to optimize the sequence of characters via CTC-loss.

In addition to the acoustic model training, the use case specific language models are trained from the generated corpora with a Kneser-Ney [21] 5-gram model. Furthermore, generic language models are also trained per language using the same technique while enabling pruning of the model. During inference time, a transcription that maximizes the $Q(y)$ as described in Equation 2 is searched.

$$Q(y) := \log(p_{ctc}(y|x)) + \alpha \log(p_{lm}(y)) + \beta \text{count}(y) \quad (2)$$

This search utilizes a linear combination of the log probabilities of the CTC trained acoustic model and the log probabilities of the provided language model with an additional word insertion counter [1]. The contribution of the language model in the result is controlled by a parameter α . The parameter β controls the insertion of additional words. Each of the control parameters are optimized during the acoustic and the language model training. Finally, beam search [8] is used to find the optimal transcription.

5.4. Glassbox environment

In the SR Lab context, the glassbox environment is a way to demonstrate the descriptive information to a user in order to outline possible design choices. A visualisation for the LVCSR module is shown in Figure 6. The first plot depicts the log-mel spectrogram of the given audio. The second one shows the possible VAD activity segments and also the most probable word assignments in the beam-search algorithm during the inference phase. The third and the fourth plot show the transcriptions obtained as selections.

5.5. Evaluation

The evaluation metric details have already been outlined in Subsection 4.4. The WER is the selected evaluation metric for the conducted experiments. Figure 7 shows the evaluation results of the LVCSR module. The x-axis indicates the target speakers during evaluation. The y-axis shows the WER score for the speaker on all of the test recordings. Microphones used in the experiments are visualized as differently colored bars to distinguish the WER results per microphone. The results demonstrate a small margin of performance change on the standard microphone in the experiments. Though, the throat microphone has a higher ratio of fluctuation and relatively lower performance. Considering the relatively low amount of available skull and throat microphone data (we only have one speaker for the skull microphone), results indicate a good level of adjustment. Therefore these results demonstrate the importance of utilizing the available data via augmentation, synthesis, or other means.

6. Conclusions and Outlook

We have described the architecture and a demonstrator of an SR Lab for the generation of tailor-fit military ASR solutions. As a proof-of-principle, we have developed two different ASR systems, namely a KWS system and an LVCSR system, for two different military use cases. Our experiments conducted on the two reported use cases have clearly shown the benefits of using a unified SR Lab for i) systematically modelling the uses cases, ii) selecting and coupling the different speech processing modules, and, iii) performance analysis of the resulting SR solution.

The focus of upcoming work needs to be on the automatic generation of training data by data augmentation and voice conversion. An additional topic is the application of transfer learning and multi-task learning as a means for addressing the scarce data problem. Further use cases to be considered are VUIs for dismounted soldiers as well as various tasks related to radio surveillance and reconnaissance (COMINT). The SR Lab is intended to enable the quick definition of such use cases and the generation of respective solutions.

References

- [1] et al, A.Y.H., 2014. Deep speech: Scaling up end-to-end speech recognition. CoRR abs/1412.5567. URL: <http://arxiv.org/abs/1412.5567>, [arXiv:1412.5567](https://arxiv.org/abs/1412.5567).
- [2] et al, D.S.P., 2019a. SpecAugment: A simple data augmentation method for automatic speech recognition, in: 20th Annual Conference of the International Speech Communication Association (Interspeech), ISCA. pp. 2613–2617.
- [3] et al, R.T., 2020. Tslern, a machine learning toolkit for time series data. *Journal of Machine Learning Research* 21, 1–6. URL: <http://jmlr.org/papers/v21/20-091.html>.
- [4] et al, V.V., 2019b. Manifold mixup: Better representations by interpolating hidden states, in: 36th International Conference on Machine Learning (ICML), PMLR. pp. 6438–6447.
- [5] Amodei, D.e.a., 2016. Deep speech 2: End-to-end speech recognition in english and mandarin, in: 33rd International Conference on International Conference on Machine Learning (ICML), JMLR.org. p. 173–182.
- [6] Bau, N., Endres, S., Gerz, M., Gökgöz, F., 2018. A cloud-based architecture for an interoperable, resilient, and scalable c2 information system, in: International Conference on Military Communication and Information Systems (ICMCIS).
- [7] Gökgöz, F., Hashem, M., 2021. Investigating the scarce data and resources problem for speech recognition using transfer learning and data augmentation, in: Elektronische Sprachsignalverarbeitung (ESSV), TUDpress. pp. 120–127.
- [8] Graves, A., 2012. Sequence transduction with recurrent neural networks. CoRR abs/1211.3711. URL: <http://arxiv.org/abs/1211.3711>, [arXiv:1211.3711](https://arxiv.org/abs/1211.3711).
- [9] Graves, A., Fernández, S., Gomez, F., Schmidhuber, J., 2006. Connectionist temporal classification: Labelling unsegmented sequence data with recurrent neural networks, in: Proceedings of the 23rd International Conference on Machine Learning (ICML), Association for Computing Machinery. p. 369–376. doi:[10.1145/1143844.1143891](https://doi.org/10.1145/1143844.1143891).
- [10] Heafield, K., 2011. KenLM: Faster and smaller language model queries, in: Proceedings of the Sixth Workshop on Statistical Machine Translation, Association for Computational Linguistics. pp. 187–197. URL: <https://aclanthology.org/W11-2123>.
- [11] Kneser, R., Ney, H., 1995. Improved backing-off for m-gram language modeling, in: International Conference on Acoustics, Speech, and Signal Processing (ICASSP), pp. 181–184. doi:[10.1109/ICASSP.1995.479394](https://doi.org/10.1109/ICASSP.1995.479394).
- [12] Lee, S., Kim, J., Lee, I., 2012. Speech/audio signal classification using spectral flux pattern recognition, in: Workshop on Signal Processing Systems, IEEE. pp. 232–236.
- [13] Ma, Y., Nishihara, A., 2013. Efficient voice activity detection algorithm using long-term spectral flatness measure. *EURASIP Journal on Audio, Speech, and Music Processing* 2013, 1–18.
- [14] Müller, M., Kurth, F., Clausen, M., 2005a. Audio matching via chroma-based statistical features, in: 6th International Conference on Music Information Retrieval (ISMIR), pp. 288–295.

- [15] Müller, M., Kurth, F., Clausen, M., 2005b. Chroma-based statistical audio features for audio matching, in: Workshop on Applications of Signal Processing to Audio and Acoustics (WASPAA), IEEE. pp. 275–278.
- [16] Petitjean, F., Ketterlin, A., Gançarski, P., 2011. A global averaging method for dynamic time warping, with applications to clustering. *Pattern recognition* 44, 678–693.
- [17] Schmitz, H.C., Cornaggia-Urrigshardt, A., Gökgöz, F., Kent, S., Wilkinghoff, K., 2019. Calm interfaces for integrated C2 systems, in: International Command and Control Research and Technology Symposium (ICCRTS).
- [18] Urrigshardt, S., Kreuzer, S., Kurth, F., 2016. General detection of speech signals in the time-frequency plane, in: ITG Symposium on Speech Communication (ITG Speech), VDE. pp. 1–5.
- [19] Von Zeddelmann, D., Kurth, F., Müller, M., 2010. Perceptual audio features for unsupervised key-phrase detection, in: International Conference on Acoustics, Speech and Signal Processing (ICASSP), IEEE. pp. 257–260.
- [20] Wilkinghoff, K., Cornaggia-Urrigshardt, A., Gökgöz, F., 2021. Two-dimensional embeddings for low-resource keyword spotting based on dynamic time warping, in: 14th ITG Conference on Speech Communication (ITG Speech), VDE-Verlag. pp. 9–13.
- [21] Zhang, H., Chiang, D., 2014. Kneser-ney smoothing on expected counts, in: 52nd Annual Meeting of the Association for Computational Linguistics (ACL), The Association for Computer Linguistics. pp. 765–774.
- [22] Zhang, H., Cisse, M., Dauphin, Y.N., Lopez-Paz, D., 2018. Mixup: Beyond empirical risk minimization, in: International Conference on Learning Representations (ICLR).
- [23] Zhang, X., Zhao, R., Qiao, Y., Wang, X., Li, H., 2019. AdaCos: Adaptively scaling cosine logits for effectively learning deep face representations, in: Conference on Computer Vision and Pattern Recognition (CVPR), IEEE. pp. 10823–10832.